Solution

360° = 24hrs

15° = 1 hour

1° = 4 minutes

Village A = 75°W

Village B = 120°E

We will add (west and east)

(75 + 120)° = 195°

195 divided by 15°

= 13hrs

Destination time = initial + calculated time

= 5:00 + 13 hrs

= 18:00

18:00 = 6:00

Answer = 6:00am on Saturday

Example 3

Calculate the local time in New York (USA) longitude 75°W, when it is 10am in Nigeria of longitude 15°E

Solution

Initial time = 10:00am

New York = 75°W

Nigeria = 15° E

We will add (west and east)

(75 + 15)° = 90°

90° divided by 15° = 6 hrs

Destination time = initial + calculated time

= 10:00am + 6hrs

= 16:00 hrs

16:00 hrs = 4:00pm

Answer = 4:00pm

GLOSSARY

1. **Dark energy:** A theoretical form of energy postulated to act in opposition to gravity and to occupy the entire universe, accounting for most of the energy in it and causing its expansion to accelerate.
2. **Magnetic field:** A force field that is created by moving electric charges and magnetic dipoles, and exerts a force on other nearby moving charges and magnetic dipoles.
3. **Penumbra:** The partially shaded outer region of the shadow cast by an opaque object.
4. **Asteroids:** Small rocky celestial bodies that revolve around the Sun, like other planets.
5. **Standard time:** A uniform time for places in approximately the same longitude, established in a country or region by law or custom.
6. **Galactic movement:** This is the movement of the earth with the sun and the rest of the solar system in an orbit around the centre of the Milky Way Galaxy.
7. **Equinox:** Time, when the apparent movement of the sun is overhead the equator, equal day and night occurs.
8. **Gibbous:** Third quarter of moon's phase is known as Gibbous.
9. **Solar flare:** A magnetic storm on the sun and releases huge amounts of gases. It can cause 'Sun quakes'.
10. **Super Nova:** The explosive death of a star. It obtains brightness of 100 million suns for a short time.

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